

ARTHUR MLAMBO

Mechatronics Engineer

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PROFILE

Mechatronics Engineer with over five years of progressive experience spanning telecom fiber infrastructure, solar energy systems, and industrial systems integration across high demand operational environments. Proven track record in commissioning, fault diagnosis, and performance monitoring of electromechanical and automated systems. Complemented by an M.Sc. in Data Science (2026) enabling advanced analytics and condition monitoring capabilities on top of core engineering expertise.

SKILLS

PLC & Control Systems: PLC Programming (Allen-Bradley), Control Systems Architecture, Electro-mechanical Systems, Ladder Logic, Function Block Diagrams, Control System Commissioning.

Network & Cabling: Structured cabling systems (fiber optic, FTTH, LAN), passive network infrastructure, fiber splicing & termination, network fault localisation, cabling design principles, rack layout & patching, signal integrity testing.

Technical Documentation: Bill of Materials (BOM) preparation, network design documentation, method statements, standard operating procedures (SOPs), commissioning reports, technical datasheets, incident trend summaries, RFP/RFQ support.

Commissioning & Fault Diagnosis: End-to-end system commissioning, fault localisation in high demand industrial environments, hardware-software interface diagnostics, performance monitoring system deployment.

Data and Analytics: Python (automation scripting, anomaly detection, monitoring pipelines), SQL, AWS (S3, Lambda, IoT data ingestion), Machine Learning for anomaly detection & predictive maintenance

Data Visualisation & Monitoring: Power BI, Tableau, Grafana, Plotly

Engineering Tools: AutoCAD, SolidWorks, MS Office (advanced), Git

WORK EXPERIENCE

Go Solar (Pvt) Ltd., Cape Town

Jan 2023 - Jun 2024

Solar Systems Engineer / Automation & Commissioning

- **System commissioning:** Commissioned and integrated 100+ industrial solar power systems end-to-end from equipment delivery and electrical installation to grid integration and performance sign off, managing the full lifecycle on each site.
- **Control & monitoring automation:** Designed and deployed automated performance monitoring systems using Python scripts interfacing with sensor data streams, applying anomaly detection to trigger fault alerts, reducing unplanned downtime across the fleet.
- **Fault diagnosis & troubleshooting:** Diagnosed and resolved electromechanical faults across distributed sites under time critical conditions, coordinating with subcontractors and grid operators to restore system availability.
- **Stakeholder management:** Coordinated cross-functional teams and external partners across multi-site rollouts, ensuring compliance with international electrical standards and client SLAs.

Frogfoot Networks, Cape Town

Oct 2018 - Dec 2020

Field Systems Technician

- Industrial infrastructure commissioning: Executed field splicing, termination, and network commissioning in high demand industrial environments, systematic verification of signal integrity from endpoint to central switching infrastructure.
- Fault localisation & troubleshooting: Diagnosed complex network and hardware faults using structured diagnostic processes, identifying root causes at the hardware software interface and restoring connectivity under operational pressure.
- Interface management: Resolved connectivity issues between distributed hardware endpoints and central control/management systems - directly analogous to BMS field device integration.

Steel Makers (Pvt) Ltd.

Jan 2014 - Dec 2014

Electrical / Mechanical Engineering Intern

- Industrial Electrical & Mechanical Systems: Supported maintenance, inspection, and troubleshooting of industrial electrical and mechanical equipment in a live steel manufacturing environment.
- HVAC & Industrial Cooling Systems: Supported the operation and maintenance of industrial cooling and HVAC-related equipment, developing practical understanding of temperature control, airflow, pumps, and mechanical cooling systems.
- PLC & Control Systems: Supported maintenance and troubleshooting of PLC controlled equipment, motor drives, relay logic, sensors, and industrial control panels.
- Controls & Monitoring: Worked at the interface between electrical hardware, sensors, control systems, and industrial software.

PROJECTS

Predictive Maintenance MLOps Pipeline

2025 – Present

Python | PyTorch | FastAPI | Docker | AWS | MLflow | SQL

- Building an end-to-end ML pipeline for fault prediction on time. series sensor data. Covers ingestion, feature engineering, training, and a FastAPI endpoint for model serving, containerised with Docker and deployed through a GitHub Actions CI/CD workflow.

M.Sc. Thesis: Hybrid ML for Predictive Maintenance

2025 – 2026

PyTorch | TensorFlow | Scikit-Learn | Pandas

- Developed a hybrid 1D-CNN + Random Forest predictive maintenance framework for industrial asset failure detection using the AI4I 2020 dataset, achieving 97.6% ROC-AUC for binary classification and 0.861 Macro F1 for multiclass fault diagnosis.

EDUCATION

M.Sc. Data Science (Distinction)

Feb 2026

Arden University, Berlin

Thesis: Hybrid ML frameworks for predictive maintenance on industrial sensor data. Modules: Machine Learning, Deep Learning, Big Data & Spark, Cloud Computing (AWS), Statistical Modelling.

B.Eng. (Hons) Mechatronics

Dec 2015

Chinhoyi University of Technology, Zimbabwe.

CERTIFICATIONS & MEMBERSHIPS

Member - Verein Deutscher Ingenieure (VDI), Berlin Chapter

PrTech - South African Institute of Engineers

LANGUAGES

English - Fluent (C2) | German - Basic (A2/B1, actively improving - based in Berlin)